

**L298N can drive ONLY 2 motors.**

To use **4 motors**, you must wire them **in pairs**:

✓ **Left side = 2 motors connected together**

✓ **Right side = 2 motors connected together**

This is the standard and correct method for 4-wheel robot cars.

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## ✓ **HOW TO CONNECT 4 MOTORS TO YOUR L298N**

### ● **Left Side Motors (2 motors)**

Take your **two left motors** (front-left + back-left):

- Left Motor 1 wire A
- Left Motor 2 wire A → **connect both together → L298N OUT1**
- Left Motor 1 wire B
- Left Motor 2 wire B → **connect both together → L298N OUT2**

✓ **OUT1 = ALL left motors forward wire**

✓ **OUT2 = ALL left motors backward wire**

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### ● **Right Side Motors (2 motors)**

Take your **two right motors** (front-right + back-right):

- Right Motor 1 wire A
- Right Motor 2 wire A → **connect both together → L298N OUT3**
- Right Motor 1 wire B
- Right Motor 2 wire B → **connect both together → L298N OUT4**

✓ **OUT3 = ALL right motors forward wire**

✓ **OUT4 = ALL right motors backward wire**

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## NOW YOUR CAR WORKS AS A 2-MOTOR SYSTEM (BUT ACTUALLY 4 MOTORS)

So the code stays **exactly same**.

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### L298N → Arduino Wiring (Same as before)

#### ### Left motors (2 motors)

- IN1 → D7
- IN2 → D6

#### Right motors (2 motors)

- IN3 → D5
- IN4 → D4

#### ENA & ENB

- Keep jumpers ON  
(Your code doesn't use PWM)
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### Battery

- Battery + → L298N +12V
  - Battery – → L298N GND
  - L298N 5V → Arduino 5V
  - L298N GND → Arduino GND
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### Ultrasonic Sensor

- TRIG → A1
  - ECHO → A2
  - VCC → 5V
  - GND → GND
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## ● Servo Motor

- Signal → D10
- VCC → 5V
- GND → GND

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## ✓ Final Summary

| Side              | No. of Motors        | Connect To  |
|-------------------|----------------------|-------------|
| Left Side Motors  | 2 motors in parallel | OUT1 & OUT2 |
| Right Side Motors | 2 motors in parallel | OUT3 & OUT4 |